

Exercise -I

Q1. A kite is flying with the string inclined at 60° to the horizon. The height of the kite above the ground, when the string is 20 mts. long, is -

- (A) 15 m (B) $15\sqrt{3}$ m (C) 10 m (D) $10\sqrt{3}$ m

Ans. D

Sol. -

Q2. If the length of the shadow of a vertical pole on the horizontal ground is equal to its height, find the angle of elevation of the sun -

- (A) 60° (B) 30° (C) 45° (D) 90°

Ans. C

Sol. -

Q3. The angle of elevation of the top of a tower from a point 20 m away from its base is 45° . The height of the tower is -

- (A) 10 m (B) 20 m (C) 40 m (D) $20\sqrt{3}$ m

Ans. B

Sol. -

Q4. If the angle of depression of a point A on the ground from the top of a tower be 30° , then the angle of elevation of the top of the tower from the point A will be-

- (A) 60° (B) 45° (C) 30° (D) None of these

Ans. C

Sol. -

Q5. At a point 15 m away from the base of a 15 m high house the angle of elevation of the top is -

- (A) 45° (B) 30° (C) 60° (D) 90°

Ans. A

Sol. -

Q6. From the top of a cliff 25 meters high, the angle of elevation of a tower is found to be equal to the angle of depression of the foot of the tower, then height of the tower is-

- (A) 25 m (B) $25\sqrt{2}$ m (C) $25/\sqrt{2}$ m (D) 50 m

Ans. D

Sol. -

Q7. Two poles of equal heights stand on either side of a 100 meters wide road. At a point between the poles, the angle of elevation of the tops of the poles are 30° and 60° . The height of each pole is -

- (A) 25 m (B) $25\sqrt{3}$ m (C) $\frac{100}{\sqrt{3}}$ m (D) None of these

Ans. B

Sol. -

Q8. A flag staff on the top of the tower 80 meter high, subtends an angle $\tan^{-1}\left(\frac{1}{9}\right)$ at a point on the ground 100 meters from the foot of the tower. Find the height of the flag-staff.

- (A) 20 m (B) 30 m (C) 25 m (D) 35 m

Ans. A

Sol. -

Q9. A person walking along a straight road observes that at two points 1 km apart, the angles of elevation of a pole in front of him are 30° and 75° . The height of the pole is,

- (A) $250(\sqrt{3} + 1)$ m (B) $250(\sqrt{3} - 1)$ m (C) $225(\sqrt{2} - 1)$ m (D) $225(\sqrt{2} + 1)$ m

Ans. A

Sol. -

Q10. An observer in a boat finds that the angle of elevation of a tower standing on the top of a cliff is 60° and that of the top of cliff is 30° . If the height of the tower be 60 meters, then the height of the cliff is -

- (A) 30 m (B) $60\sqrt{3}$ m (C) $20\sqrt{3}$ m (D) None of these

Ans. A

Sol. -

Q11. A tree is broken by wind, its upper part touches the ground at a point 10 meters from the foot of the tree and makes an angle of 45° with the ground. The entire length of the tree is -

- (A) 15 meters (B) 20 meters (C) $10(1 - \sqrt{2})$ meters (D) $10(1 + \sqrt{3}/2)$ meters

Ans. C

Sol. -

Q12. The length of the shadow of a pole inclined at 10° to the vertical towards the sun is 2.05 meters, when the elevation of the sun is 38° . The length of the pole is -

- (A) $\frac{2.05 \sin 38^\circ}{\sin 42^\circ}$ (B) $\frac{2.05 \sin 42^\circ}{\sin 38^\circ}$ (C) $\frac{2.05 \cos 38^\circ}{\cos 42^\circ}$ (D) None of these

Ans. A

Sol. -

Q13. At a distance $2h$ from the foot of a tower of height h , the tower and a pole at the top of the tower subtend equal angles. Height of the pole should be -

- (A) $\frac{5h}{3}$ (B) $\frac{4h}{3}$ (C) $\frac{7h}{5}$ (D) $\frac{3h}{2}$

Ans. A

Sol. -

Q14. A flag-staff stands upon the top of a building. If at a distance of 40 meters from the base of the building, the angles of elevation of the topes of the flag-staff and the building are 60° and 30° respectively, then the height of the flag - staff is -

- (A) 46.17 m (B) 50 m (C) 25 m (D) None of these

Ans. A

Sol. -

Q15. A tower subtends an angle α at a point A in the plane of its base and the angle of depression of the foot of the tower at a point ℓ meters just above A is β . The height of the tower is -

- (A) $\ell \tan \beta \cot \alpha$ (B) $\ell \tan \alpha \cot \beta$ (C) $\ell \tan \alpha \tan \beta$ (D) $\ell \cot \alpha \cot \beta$

Ans. B

Sol. -

Q16. The angle of elevation of the top of a hill from each of the vertices A, B, C of a horizontal triangle is a . The height of the hill is -

- (A) $b \tan \alpha \cdot \operatorname{cosec} B$ (B) $\frac{1}{2} \tan \alpha \cdot \operatorname{cosec} A$ (C) $\frac{1}{2} c \tan \alpha \cdot \operatorname{cosec} C$

(D) None of these

Ans. B, C

Sol. -

Q17. The angle of elevation of the top of a TV tower from three points A, B, C in a straight line (in the horizontal plane) through the foot of the tower are α , 2α , 3α respectively. If

AB = α , the height of the tower is -

- (A) $a \tan \alpha$ (B) $a \sin \alpha$ (C) $a \sin 2\alpha$ (D) $a \sin 3\alpha$

Ans. C

Sol. -

Q18. ABCD is a square plot. The angle of elevation of the top of a pole standing at D from A or C is 30° and that from B is θ , then $\tan \theta$ is equal to -

- (A) $\sqrt{6}$ (B) $1/\sqrt{6}$ (C) $\sqrt{3}/\sqrt{2}$ (D) $\sqrt{2}/\sqrt{3}$

Ans. B

Sol. -

Q19. The angle of elevation of the top of the tower observed from each of the three points A, B, C on the ground, forming a triangle is the same angle α . If R is the circum-radius of the triangle ABC, then the height of the tower is -

- (A) $R \sin \alpha$ (B) $R \cos \alpha$ (C) $R \cot \alpha$ (D) $R \tan \alpha$

Ans. D

Sol. -

Q20. A house of height 100 meters subtends a right angle at the window of an opposite house. If the height of the window be 64 meter, then the distance between the two houses is -

- (A) 48 m (B) 36 m (C) 54 m (D) 72 m

Ans. A

Sol. -

Q21. On the level ground the angle of elevation of the top of a tower is 30° . On moving 20 mt. nearer the tower, the angle of elevation is found to be 60° . The height of the tower is-

- (A) 10 m (B) 20 m (C) $10\sqrt{3}$ m (D) 15 m

Ans. C

Sol. -

Q22. An observer on the top of a tree, finds, the angle of depression of a car moving towards the tree to be 30° . After 3 minutes this angle becomes 60° . After how much more time, the car will reach the tree -

- (A) 4 min. (B) 4.5 min. (C) 1.5 min. (D) 2 min.

Ans. C

Sol. -

Q23. The angles of elevation of the top of a tower from two point A and B on the ground distant a and b from the tower are complimentary. If the line AB passes through the foot of the tower, the height of the tower is -

- (A) ab (B) $\frac{a}{b}$ (C) $\sqrt{ab}$ (D) $\sqrt{\frac{a}{b}}$

Ans. C

Sol. -

Q24. A vertical lamppost of height 9 metres stands at the corner of a rectangular field. The angle of elevation of its top from the farthest corner is 30° , while from another corner it is 45° . The area of the field is -

- (A) $81\sqrt{2}$ metre² (B) $9\sqrt{2}$ metre² (C) $81\sqrt{3}$ metre² (D) $9\sqrt{3}$ metre²

Ans. A

Sol. -

Q25. A flagstaff stands vertically on a pillar, the height of the flagstaff being double the height of the pillar. A man on the ground at a distance finds that both the pillar and the flagstaff subtend equal angles at his eyes. The ratio of the height of the pillar and the distance of the man from the pillar, is -

- (A) $\sqrt{3} : 1$ (B) $1 : 3$ (C) $1 : \sqrt{3}$ (D) $\sqrt{3} : 2$

Ans. C

Sol. -

Q26. The upper $\frac{3}{4}$ th portion of a vertical pole subtends an angle $\tan^{-1} \frac{3}{5}$ at a point in the horizontal plane through its foot and at a distance 40 m from the foot. A possible height of the vertical pole is -

- (A) 80m (B) 20m (C) 40 m (D) 60 m

Ans. C

Sol. -

Q27. A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank of the river is 60° and when he retires 40 meters away from the tree the angle of elevation becomes 30° . The breadth of the river is -

- (A) 20 m (B) 30 m (C) 40 m (D) 60 m

Ans. A

Sol. -

Q28. A tower stands at the centre of a circular park. A and B are two points on the boundary of the park such that AB (= a) subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30° . The height of the tower is -

- (A) $2a/\sqrt{3}$ (B) $2a\sqrt{3}$ (C) $a/\sqrt{3}$ (D) $a\sqrt{3}$

Ans. C

Sol. -

Q29. AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation of the point A from a certain point C on the ground is 60° . He moves away from the pole along the line BC to a point D such that CD = 7 m. From D the angle of elevation of the points A is 45° . Then the height of the pole is

- (A) $\frac{7\sqrt{3}}{2}(\sqrt{3} + 1)m$ (B) $\frac{7\sqrt{3}}{2}(\sqrt{3} - 1)m$ (C) $\frac{7\sqrt{3}}{2} \frac{1}{\sqrt{3} + 1}m$ (D) $\frac{7\sqrt{3}}{2} \frac{1}{\sqrt{3} - 1}m$

Ans. A

Sol. -

Q30. A bird is sitting on the top of a vertical pole 20 m high and its elevation from a point O on the ground is 45° . It flies off horizontally straight away from the point O. After one second, the elevation of the bird from O is reduced to 30° . Then the speed (in m/s) of the bird is :

- (A) $20\sqrt{2}$ (B) $20(\sqrt{3} - 1)$ (C) $40(\sqrt{2} - 1)$ (D) $40(\sqrt{3} - \sqrt{2})$

Ans. B

Sol. -

Q31. If the angles of elevation of the top of a tower from three collinear points A, B and C on a line leading to the foot of the tower, are 30° , 45° and 60° respectively, then the ratio, AB : BC is :

- (A) $\sqrt{3} : 1$ (B) $\sqrt{3} : \sqrt{2}$ (C) $1 : \sqrt{3}$ (D) $2 : 3$

Ans. A

Sol. -

Q32. A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30° . After walking for 10 minutes from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60° . Then the time taken (in minutes) by him, from B to reach the pillar, is :

- (A) 20 (B) 5 (C) 6 (D) 10

Ans. B

Sol. -

Q33. Let a vertical tower AB have its end A on the level ground. Let C be the mid-point of AB & P be a point on the ground such that $AP=2AB$. If $\angle BPC = \beta$, then $\tan \beta$ is equal to

- (A) $\frac{2}{9}$ (B) $\frac{4}{9}$ (C) $\frac{6}{7}$ (D) $\frac{1}{4}$

Ans. A

Sol. -

Q34. PQR is a triangular park with $PQ = PR = 200\text{m}$. A T.V. tower stands at the mid-point of QR. If the angles of elevation of the top of the tower at P, Q and R are respectively 45° and 30° and 30° , then the height of the tower (in m) is

- (A) $100\sqrt{3}$ (B) $50\sqrt{2}$ (C) 100 (D) 50

Ans. C

Sol. -